Project Plan

Radiation-Induced Defect Evolution and Device Degradation in Silicon

Single living roadmap for project architecture, workflow, file map, and module sequencing

Purpose. This document is the single living roadmap for **Radiation Effects Project 2**. The top-level `README.md`, `project_plan.tex`, and `project_plan.pdf` are the maintained architecture documents and should be updated whenever the project architecture changes.

Contents

Current architecture revision. Module 1 is now planned as an **interaction-resolved Geant4 campaign** with separate photon and neutron runs across targeted energy bands so that different microscopic interaction mechanisms can be studied intentionally rather than mixed together in one broad source case. Modules 2–6 are now 2D-first. Module 2 now includes a first executable two-dimensional finite-element Poisson solver using linear triangular elements for defect-dependent space charge. Module 3 includes a linear triangular finite-element diffusion-reaction path with mass, diffusion, and reaction matrix assembly, in parallel with the earlier structured-grid implementation. Module 3 is now the preferred project base case for causal physics-informed neural-network training because its defect concentration is governed by a genuine time-dependent initial-value problem. The `module3_cPIN.tex/.pdf` note develops that method from an introductory level, and `main_module3_CPINN_Defect_Evolution.m` now implements the first scalar constant-coefficient benchmark with dimensionless residuals, ordered physical-time slices, stop-gradient exponential weights, analytic/FEM references, and causal diagnostics. The existing Module 2 causal-PIN files remain in the project as an ordered quasi-static electrostatic continuation study. Module 4 is now a single **two-dimensional ballistic-diffusive thermal-transport module** adapted from the ballistic-diffusive heat-conduction framework developed in the FEM_BallisticDiffusion_PINN project. Module 4 now includes both the first structured-grid ballistic-diffusive path and a linear triangular FEM path with heat-capacity matrix, thermal-conductivity matrix, source-vector assembly, and backward-Euler time stepping. Module 5 now includes a first linear triangular FEM drift-diffusion path with carrier mass, diffusion, electric-field drift, and linear recombination matrix assembly. Module 6 now includes a first staggered linear triangular FEM coupling scaffold that places defect concentration, electrostatic potential, temperature, electron density, and hole density on a shared mesh and iterates the coupled maps. The older Module 4a/4b thermal notes are archived as legacy baseline material. Module 7 remains the scaling and multiscale-prediction module, Module 8 records reduced-fidelity and importance-scoring logic, Module 9 now defines the transmon geometry, microwave-package context, and reduced 2D FEM domains for the superconducting-device extension. Module 10 supplies the BCS, Ginzburg–Landau, and time-dependent Ginzburg–Landau theory foundation. Module 11 defines a physics-constrained graph-learning program for recovering the directed physical dependencies among Module 2–5 variables, classifying mechanism type, strength, sign, and lag, identifying feedback groups, and deriving a physically admissible partial coupling sequence from synthetic trajectories and interventions.

1 Project objective

The project builds a physics-based framework for connecting a radiation event in silicon to the eventual degradation of device performance. The goal is not only to model where energy is deposited, but also to understand how that deposited energy creates defects, how those defects evolve with time and temperature, how they alter the internal electric fields and carrier transport, and ultimately how they change the behavior of the device.

The intended physics chain is:

1. incident radiation deposits energy in silicon,
2. deposited energy seeds point defects and effective defect populations,
3. defects migrate, react, cluster, dissociate, and anneal,
4. charged defects alter the internal electrostatic landscape,
5. thermal evolution changes defect kinetics and transport coefficients,
6. the changed field and trap population alter carrier transport,
7. the coupled system produces measurable device degradation.

2 Module roadmap

No.	Module	Purpose
1	Interaction-resolved Geant4 radiation deposition campaign in silicon	Run separate photon and neutron simulations in targeted energy bands so that distinct interaction mechanisms can be isolated, compared, and translated into deposition maps, secondary-particle spectra, and damage-precursor metrics for later modules.
2	Two-dimensional electrostatics with defect-dependent space charge	Solve the 2D electrostatic Poisson problem for a damaged silicon structure using a linear triangular finite-element implementation. Convert dopants and charged defects into electric potential, electric field, and depletion-structure changes.
3	Two-dimensional defect evolution with material-aware kinetic coefficients	Evolve vacancies, interstitials, complexes, and effective defect species in 2D using diffusion, reaction, migration, and annealing laws whose reduced coefficients can depend on temperature, charge state, dopants, and electric field. The implementation includes structured-grid and linear-triangle FEM paths plus a first causal-PINN driver for the scalar constant-coefficient transient equation. The detailed causal-PIN note and executable pure-annealing benchmark make Module 3 the preferred time-dependent case for causality-respecting ordered residual weighting.

4	Two-dimensional ballistic-diffusive thermal transport in silicon	Use a reduced ballistic-diffusive heat-conduction framework to evolve the 2D thermal field in silicon, retain finite-flight-time and boundary-memory effects beyond Fourier conduction, support volumetric radiation-related heat sources, and update temperature-dependent defect and device coefficients. The implementation now includes a structured-grid path and a first linear-triangle FEM path for the ballistic-diffusive thermal equation.
5	Two-dimensional drift-diffusion carrier transport with defect-assisted recombination	Predict how the damaged device transports electrons and holes under the altered field, temperature, and trap landscape. The implementation now includes a first linear-triangle FEM path with mass, diffusion, drift, and linearized recombination matrix assembly.
6	Coupled two-dimensional multiphysics degradation model	Couple Modules 2, 3, 4, and 5 into a self-consistent simulator for defect evolution, field evolution, thermal evolution, and carrier-transport degradation. The current implementation adds a first staggered shared-mesh linear-triangle FEM coupling scaffold with convergence and charge-consistency diagnostics.
7	Multiscale extrapolation and scalable prediction	Develop methods for making predictions on much larger spatial, temporal, or statistical scales than the direct high-fidelity simulation can afford. This module covers controlled approximations, quantified error bars, statistical upscaling, and hybrid multiresolution workflows.
8	Reduced-fidelity and importance-based physics selection	Rank physics components by output impact per computational cost, define reduced-model error metrics, and guide which full-physics pathways should be retained in practical sweeps.
9	Transmon geometry, microwave package, and 2D FEM domains	Define the superconducting transmon-like device geometry, including the microwave-aware physical package picture and the reduced two-dimensional MATLAB FEM target with substrate, pads, leads, effective JJ region, normal-metal trap, backside heat sink, bond pads, and wiring boundary treatment.
10	BCS, Ginzburg–Landau, and time-dependent Ginzburg–Landau superconductivity theory	Establish the microscopic and phenomenological theory required to interpret superconducting order, quasiparticles, Josephson coupling, spatial order-parameter variation, and nonequilibrium superconducting dynamics in the device branch.

11	Graph neural networks for physical dependency discovery	Recover directed physical-variable edges, relation types, strengths, signs, and time lags from graph-family synthetic trajectories and matched interventions; identify strongly connected feedback groups and derive a partial physical coupling sequence with physics constraints and uncertainty.

3 Project narrative and module descriptions

Radiation-Induced Defect Evolution and Device Degradation in Silicon

This project builds a physics-based framework for connecting a radiation event in silicon to the eventual degradation of device performance. The goal is not only to model where energy is deposited, but also to understand how that deposited energy creates defects, how those defects evolve with time and temperature, how they alter the internal electric fields and carrier transport, and ultimately how they change the behavior of the device. Each module isolates one layer of the physics, while the final module combines them into a self-consistent multiphysics model.

Module 1. Interaction-Resolved Geant4 Radiation Deposition Campaign in Silicon

This module provides the radiation input to the rest of the framework, but it now does so through a structured set of interaction-targeted simulation campaigns rather than a single broad source run. Using Geant4, the simulation tracks incoming radiation and calculates where, when, and how much energy is deposited inside the silicon while also recording which physical interaction mechanisms dominate in each energy band. The output of this step represents the initial physical insult to the material. In practice, this module gives the spatial and possibly temporal distribution of deposited energy, secondary-particle production, and interaction-specific deposition signatures that can later be translated into an initial defect-generation source term or damage map.

For photons, the current planning assumption is to split the campaign into bands chosen to emphasize Rayleigh scattering, the photoelectric effect, Compton scattering, pair production, photonuclear interactions in the resonance region, and deep-inelastic or very-high-energy behavior. The approximate photon planning bands currently under consideration are eV to about 100 keV for Rayleigh-dominated studies, eV to about 500 keV for photoelectric-dominated studies, roughly 10 keV to 10 MeV for Compton-dominated studies, above 1.022 MeV for pair-production access, roughly 5 MeV to a few GeV for photonuclear studies, and GeV-scale and above for deep-inelastic or high-energy cascade studies. These bands are not interpreted as sharp exclusive boundaries; they are campaign ranges chosen to emphasize different interaction families.

For neutrons, the same philosophy will be used: separate runs will be designed to target different interaction regimes rather than mixing all neutron energies into one undifferentiated source case. The exact neutron energy partition and the dominant reactions to emphasize will be finalized as the

Module 1 campaign plan is expanded. In all cases, Module 1 should save not only deposited-energy maps but also interaction tallies, secondary-particle information, and compact diagnostics that explain why one source case produces a different downstream defect precursor than another.

Module 2. Electrostatics with Defect-Dependent Space Charge in a 2D Silicon Device Cross Section

This module models how radiation-induced defects modify the electrostatic state of the device in two spatial dimensions. Defects can act as charged centers, traps, or ionized states that contribute to the net space-charge density. That altered space charge changes the electric potential and electric field inside the device. Physically, this matters because the internal field controls depletion behavior, carrier motion, and ultimately how the device responds under bias. This module therefore connects the evolving defect population to the device's internal field structure.

The current implementation solves the scalar Poisson equation with a Galerkin finite-element method on a rectangular triangular mesh. The nodal unknown is electrostatic potential, the element matrix is assembled from $\int \varepsilon_{\mathrm{Si}} \nabla N_a \cdot \nabla N_b \, d\Omega$, the source vector is assembled from $\int N_a \rho \, d\Omega$, Dirichlet contact potentials are imposed strongly, and homogeneous Neumann boundaries enter naturally. The first verification cases check the zero-charge limit, linear Laplace solution, uniform space-charge limit, and localized charged-defect perturbation.

Module 3. Two-Dimensional Diffusion-Reaction Solver for Defect Migration and Annealing

This module describes the time evolution of radiation-induced defects after they are created. Defects are not static: they can diffuse, recombine, cluster, dissociate, or anneal depending on temperature and local conditions. In the upgraded Module 3 plan, the diffusion and annealing coefficients are treated as material-aware reduced kinetic coefficients rather than as universal constants. Their microscopic origin lies in bonding, lattice geometry, and electronic structure, while the first implemented simulation-level dependencies are temperature, defect charge state, dopant environment, and electric field. A 2D diffusion-reaction model then evolves the resulting defect concentrations in space and time. This is one of the core physics modules of the project because it determines how initial damage develops into a changing defect landscape, rather than treating radiation damage as frozen in place.

The `docs/physics_notes/module3_cPIN.tex/.pdf` note makes this transient equation the preferred project demonstration for causal physics-informed neural networks. It derives ordinary PINN residual, initial-condition, boundary, and optional FEM-data losses; explains the causal failure mode of simultaneous global-in-time residual minimization; and introduces ordered time-slice weights with stop-gradient treatment. The first executable MATLAB driver now implements that residual-only causal formulation for scalar constant coefficients, homogeneous zero-flux boundaries, pure annealing, Gaussian diffusion, and uniform-state preservation. The structured-grid and FEM solvers remain independent reference trajectories.

Module 4. Two-Dimensional Ballistic-Diffusive Thermal Transport in Silicon

This module adds temperature as an active physical driver using a reduced ballistic-diffusive heat-conduction model derived from the carrier Boltzmann picture. The purpose is to retain a clearer microscopic foundation than a plain Fourier heat equation while still keeping the solver executable on the same 2D device cross section used by the other modules. The thermal field controls defect diffusivities, annealing rates, and later carrier-transport coefficients. In the revised architecture, Module 4 accepts volumetric source terms appropriate for radiation-related heating and retains an explicit ballistic contribution through a first reduced rectangular-domain boundary-emission closure so that finite-flight-time and boundary-memory effects can already be represented in the initial implementation. The new FEM path derives and implements the weak form using linear triangular elements, a consistent heat-capacity matrix, a thermal-conductivity matrix, a distributed source vector, natural no-flux boundaries, and backward-Euler time stepping for the second-order-in-time relaxation equation.

Module 5. Two-Dimensional Drift-Diffusion Carrier Transport with Defect-Assisted Recombination

This module translates material damage into electrical performance changes. Using 2D drift-diffusion transport equations, the model solves for electron and hole motion under the influence of the internal electric field and carrier concentration gradients. Radiation-induced defects enter through trap-assisted recombination, trapping, lifetime degradation, and defect-limited mobility. The new FEM path expresses the electron and hole equations in weak form, stores carrier concentrations as nodal unknowns on a triangular mesh, assembles carrier mass, diffusion, electric-field drift, and linear recombination matrices, and advances the reduced equations with backward-Euler time stepping. In other words, this module shows how the evolving defect population, altered electrostatics, and temperature field affect measurable device behavior such as leakage current, charge collection, recombination losses, or degraded diode response.

Module 6. Combined Multiphysics Model Using Modules 2, 3, 4, and 5

This module integrates the previous physics pieces into a coupled framework. The defect population changes the space charge; the space charge changes the electric field; the electric field influences carrier transport; the temperature affects defect migration and annealing; and carrier behavior and local power dissipation can in turn interact with the thermal state. This module is where the project becomes a true device-degradation simulator rather than a collection of separate models. Its purpose is to produce a self-consistent prediction of how radiation exposure evolves into long-term electrical degradation.

Module 7. Multiscale Extrapolation and Scalable Prediction

This module addresses the fact that direct high-fidelity simulations may be too expensive for large devices, long times, broad parameter sweeps, or population-level reliability predictions. It covers controlled reduced-fidelity approximations, statistical upscaling, and hybrid multiresolution workflows. A central requirement is that any speedup or approximation be accompanied by a quantitative error metric relative to a higher-fidelity reference.

Module 8. Reduced-Fidelity and Importance-Based Physics Selection

This module formalizes how the project selects which physics pathways are worth paying for computationally. It defines reduced-model error in output space, ablation-based impact measures, computational-cost estimates, and impact-per-cost importance scores. It supports practical decisions about when to keep or omit specific field solves, coupling terms, defect species, or fidelity upgrades.

Module 9. Transmon Geometry, Microwave Package, and 2D FEM Domains

This module defines the first qubit-relevant superconducting geometry for the project. It records the full microwave-aware transmon package picture, including CPW resonator, coupling capacitor, bond pads, wire bonds, normal-metal quasiparticle trap, and backside heat sink. It also defines the reduced 2D front cross-section selected for MATLAB FEM implementation. The first coding target will encode the substrate, aluminum pads, junction leads, effective JJ region, normal-metal trap, backside heat sink or sink boundary, and bond pads, while treating above-surface wire-bond arcs as boundary/source terms rather than meshed domains.

Module 10. BCS, Ginzburg–Landau, and Time-Dependent Ginzburg–Landau Theory

This module supplies the superconductivity theory foundation for the device branch. It connects the normal-state electron gas, pairing interaction, BCS ground state, quasiparticle excitation spectrum, gap equation, condensate phase, Ginzburg–Landau free energy, coherence and penetration lengths, flux quantization, Josephson relations, and time-dependent order-parameter evolution. It is a theory module rather than an independent numerical block in the present Module 6 semiconductor loop.

Module 11. Graph Neural Networks for Physical Dependency Discovery

This module asks whether graph neural networks and graph-structure-learning methods can recover the directed physical dependency graph connecting defect, electrostatic, carrier, and thermal quantities in Modules 2–5. The proof of concept will generate multiple graph families using explicit mechanism switches and continuous edge multipliers, produce matched interventions, infer direct versus indirect edges, classify relation type and lag, quantify state-dependent edge strength, and identify strongly connected feedback groups. A physical coupling sequence is derived by collapsing feedback groups into a condensation graph and topologically ordering that graph. Spatial mesh adjacency and numerical solver scheduling are explicitly excluded from the Module 11 learning target.

4 Module 3 kinetic-coefficient hierarchy

Module 3 now treats the defect diffusivity D and annealing coefficient k_{ann} as reduced constitutive quantities whose values are shaped by underlying material physics. The project explicitly separates two levels of description.

At the microscopic level, defect motion and defect removal are controlled by interatomic bonding, local lattice geometry, and electronic structure. Those factors determine migration barriers, reaction barriers, jump distances, charge-state energetics, and attempt frequencies. They are the physical origin of the Arrhenius-type activation energies and prefactors used in the continuum model.

At the simulation level, Module 3 promotes the coefficients from simple functions of temperature to state-dependent reduced laws. The first implemented dependence set is

$$D = D\left(T, q_d, N_{\mathrm{dop}}, |\mathbf{E}|\right), \tag{4.1}$$

and

$$k_{\mathrm{ann}} = k_{\mathrm{ann}}\left(T, q_d, N_{\mathrm{dop}}, |\mathbf{E}|\right), \tag{4.2}$$

where q_d is the defect charge state, N_{dop} is a dopant-environment measure, and $\mathbf{E}$ is the device-scale electric field imported from Module 2 when that coupling is enabled. Strain and interface proximity are planned as the next refinement layer, while bonding and electronic structure remain the microscopic origin of the reduced parameters rather than direct continuum fields.

This hierarchy keeps the project executable while still making clear which physical effects actually contribute to the effective kinetic coefficients.

5 Module 1 interaction-resolved campaign plan

Module 1 is now organized as an interaction-resolved Geant4 campaign rather than a single undifferentiated radiation-deposition run. The purpose is to select source energies deliberately so that different microscopic interaction mechanisms can be emphasized, compared, and interpreted in downstream damage modeling.

5.1 Why this change matters

Different interaction mechanisms can produce very different spatial deposition patterns, secondary-particle populations, and damage precursors even when the total incident energy is similar. Separating the runs by targeted interaction regime makes it easier to explain why one source case leads to dense local energy deposition, another to broader electron-mediated energy transfer, and another to recoil nuclei or higher-energy secondaries that may matter more for defect production.

5.2 Photon campaign structure

Interaction emphasis	Approximate planning range	Reason for separate campaign
Rayleigh scattering	eV to about 100 keV	Isolate elastic scattering and angular redistribution with relatively small direct energy transfer per event.
Photoelectric effect	eV to about 500 keV	Study strong local energy absorption and dense short-range secondary-electron production.
Compton scattering	about 10 keV to about 10 MeV	Study partial-energy transfer, recoil-electron generation, and broader distributed deposition.
Pair production	above 1.022 MeV	Study electron-positron creation and more extended shower-like deposition behavior.
Photonuclear or resonance-region interactions	about 5 MeV to about 3 GeV	Study photonuclear reactions, recoil nuclei, and higher-energy secondaries that can change defect-precursor character.
Deep-inelastic or very-high-energy interactions	GeV scale and above	Study high-energy cascades and hadronic secondary production in the extreme-energy regime.

These are campaign ranges rather than sharp exclusive boundaries. Multiple processes can contribute in overlapping regions, but the ranges are still useful because they bias the simulation toward distinct physical mechanisms and signatures.

5.3 Neutron campaign structure

Neutrons should be treated with the same philosophy: use separate campaign families to emphasize different reaction classes and energy regimes rather than combining all neutron energies into one broad source case. The exact neutron bands are left as a planned design task so that they can be chosen deliberately around the reactions that matter most for silicon damage, displacement production, recoil spectra, and secondary charged-particle generation.

5.4 Recommended Module 1 outputs

Each interaction-resolved Module 1 run should save, at minimum, a deposited-energy map, depth and lateral deposition summaries, interaction-type tallies, secondary-particle spectra or counts when available, and metadata describing the incident particle, targeted energy band, physics list, geometry revision, binning, and units. This turns Module 1 into both a source generator and an interpretation layer for downstream defect modeling.

6 Core architecture decision: 2D for Modules 2–6

The original one-dimensional path was numerically attractive, but it suppresses several effects that become important once the damage pattern is spatially structured: lateral spreading of defect populations, lateral field bending, nonuniform temperature fields near localized energy deposition, and current crowding or carrier-flow rerouting. A 2D architecture is therefore the more faithful bridge between a realistic Geant4 source pattern and later electrostatic, thermal, and transport calculations.

This move to 2D does not mean every solver should start maximally complex. The first versions should still keep simple rectangular domains, structured grids, small defect networks, simple boundary conditions, and weak one-way coupling before self-consistent feedback.

7 Run-order workflow for the project

The run order below is the logical flow of a coupled study.

Step 1. Module 1: Geant4 radiation transport
Score deposited energy, hit coordinates, timing, interactions, and damage-source products.

Step 2. Geant4-to-MATLAB handoff
Map Geant4 output to a 2D deposition or initial defect-source field.

Step 3. Module 3: defect evolution
Evolve defects under diffusion, reactions, annealing, and source terms.

Step 4. Module 4: ballistic-diffusive thermal field
Update the ballistic-diffusive temperature field and temperature-dependent coefficients.

Step 5. Module 2: electrostatics
Build space charge and solve for potential and electric field.

Step 6. Module 5: carrier transport
Solve electron and hole drift, diffusion, recombination, and trapping.

Step 7. Module 6: coupled iteration
Iterate Modules 2–5 to a self-consistent coupled state.

Step 8. Module 11: physical graph study
Generate graph-labeled trajectories and recover directed physical dependencies, lags, and feedback groups.

Step 9. Module 7: scalable prediction
Extend predictions using reduced-order, statistical, or multiresolution methods.

Important distinction. The revised Module 4 is not a plain Fourier baseline. It is a reduced ballistic-diffusive thermal model chosen because it preserves more of the microscopic transport physics while remaining executable in 2D. In simplified studies, thermal information does not always have to be solved before electrostatics, but in the intended coupled workflow the temperature field should be available before the tightly coupled 2D solve because temperature changes diffusion, annealing, mobility, and recombination parameters.

8 Recommended coding order

1. **Module 1:** Geant4 geometry, scoring, and export format.
2. **Module 3:** run the implemented pure-annealing causal-PINN regression benchmark, verify convergence with a production training budget, then add the matched ordinary-PINN and Gaussian-diffusion comparison.
3. **Module 2:** 2D electrostatics with prescribed charged defects; first FEM Poisson path now

implemented and ready for coupling to Module 3 defect fields.

4. **Module 4:** 2D ballistic-diffusive thermal solver plus temperature-dependent coefficient handoff to later modules.
5. **Module 5:** 2D drift-diffusion with prescribed electrostatic field, thermal field, and defect/trap fields; first linear-triangle FEM path now implemented.
6. **Module 6:** weak coupling, then stronger feedback and iterative convergence.
7. **Module 7/8:** scalable-prediction methods and impact-per-cost physics selection after at least one credible high-fidelity coupled workflow exists.
8. **Module 9:** transmon geometry and 2D FEM domain tagging for the superconducting extension.
9. **Module 10:** use the BCS/GL/TDGL note as the theory basis for superconducting constitutive models and device metrics.
10. **Module 11:** define the candidate physical-variable graph, generate multiple graph families and matched interventions, establish non-graph structure-learning baselines, then train a typed GNN for directed edge, relation, strength, lag, and feedback-group recovery.

9 File-map overview

Path	**Role**
`README.md`	Public repository summary and high-level entry point.
`project_plan.tex/.pdf`	Single living roadmap document: project objective, module definitions, run order, file map, interfaces, and roadmap updates.
`docs/physics_notes/`	One large physics note per module, with derivations, symbol definitions, assumptions, and physical interpretation.
`docs/physics_notes/module3_cPIN.tex/.pdf`	Introductory causal-PIN tutorial, implemented first MATLAB benchmark, and extension design for temporally ordered defect diffusion-reaction training and validation.
`docs/implementation_notes/`	Module-by-module implementation specifications, solver notes, and interface notes.
`geant4/`	Geant4-side setup, interaction-resolved campaign macros, and example outputs for Module 1.

`matlab/`	MATLAB drivers, source code, tests, outputs, and cross-module interfaces for Modules 2–7; Module 9 supplies the next geometry extension and Module 11 supplies the planned physical-dependency graph data and learning workflow.

9.1 Current MATLAB map plus planned 2D emphasis

Path	**Role in the updated 2D plan**
`matlab/main_module2_electrostatics.m`	Top-level entry for the 2D electrostatic solve.
`matlab/main_module3_2d_defect_evolution.m`	Top-level entry for the structured-grid 2D defect-evolution solve.
`matlab/main_module3_fem_defect_evolution.m`	Top-level entry for the linear triangular FEM defect diffusion-reaction solve.
`matlab/main_module3_CPINN_Defect_Evolution.m`	Top-level causal-PINN driver with dimensionless automatic-differentiation residuals, ordered physical-time slices, stop-gradient causal weights, analytic/FEM references, and training diagnostics.
`matlab/main_module4_2d_ballistic_diffusive_thermal.m`	Top-level entry for the structured-grid Module 4 two-dimensional ballistic-diffusive thermal solve.
`matlab/main_module4_fem_ballistic_diffusive_thermal.m`	Top-level entry for the linear triangular FEM Module 4 ballistic-diffusive thermal solve.
`matlab/main_module5_drift_diffusion.m`	Top-level entry for the 2D FEM carrier-transport solve.
`matlab/main_module5_fem_drift_diffusion.m`	Compatibility wrapper for the Module 5 linear triangular FEM carrier solver.
`matlab/main_module6_multiphysics.m`	Top-level coupled FEM driver that orchestrates Modules 2, 3, 4, and 5 through a staggered Picard-style iteration.
`matlab/main_module11_generate_dependency_dataset.m`	Planned Module 11 driver for graph-family sampling, constrained physical trajectories, matched interventions, graph labels, and dataset export.
`matlab/main_module11_train_dependency_gnn.m`	Planned Module 11 driver for structure-learning baseline comparison, directed typed graph-model training, calibration, and held-out evaluation.

`matlab/src/module11_gnn/`	Planned variable-graph construction, graph-family sampling, intervention generation, feature extraction, directed edge/type/lag losses, SCC sequencing, uncertainty, and evaluation utilities.
`matlab/src/grid/`	Shared 2D grid generation and spatial operators.
`matlab/src/module3_defects/`	Module 3 defect evolution operators, initialization helpers, material-aware coefficients, and the new FEM mass/diffusion/reaction assembly functions.
`matlab/src/module4_thermal/`	Module 4 ballistic-diffusive thermal operators, initialization helpers, source-term assembly, reduced ballistic-flux closures, FEM heat-capacity and conductivity assembly, implicit FEM stepping, and verification support.
`matlab/src/module2_electrostatics/`	Implemented Module 2 finite-element Poisson operators, triangular mesh generation, charge assembly, boundary-condition insertion, and field postprocessing.
`matlab/src/module5_transport/`	Module 5 FEM drift-diffusion operators, carrier matrix assembly, linear recombination closure, mobility/source evaluation, current extraction, plotting, and summary output.
`matlab/src/diagnostics/`	Shared diagnostics, error metrics, and summary writers.
`matlab/src/io/`	Shared I/O helpers and history save utilities.
`matlab/src/plot/`	Shared 2D visualization and history plotting utilities.
`matlab/tests/`	Verification problems and regression tests, including Module 2 electrostatic limiting-case checks, the Module 3 causal-PINN pure-annealing regression/physics test, Module 3 FEM diffusion-reaction checks, Module 4 FEM thermal checks, and Module 5 FEM carrier-transport checks.
`matlab/outputs/`	Saved run products, plots, summaries, and histories.

10 Thermal-model rationale and comparison strategy

10.1 Why the ballistic-diffusive model is now primary

The revised Module 4 becomes the primary thermal branch because it preserves a clearer microscopic connection to finite-speed heat-carrier transport than the old continuum baseline while still remaining executable in 2D. The model is more ambitious than Fourier conduction, but it is still substantially simpler than a full angularly resolved phonon Boltzmann solve. The added FEM path makes the module easier to defend numerically because the thermal equation is now expressed through a weak form, triangular shape functions, element matrices, and assembled global time-stepping equations.

10.2 How it should still be benchmarked

Even though the split 4a/4b architecture is removed, it is still useful to compare the new Module 4 against a simple Fourier benchmark when validating the implementation. Useful metrics include

- peak temperature and hot-spot width,
- early-time surface or boundary heat flux,
- lateral spreading rates of a localized thermal pulse,
- annealing-rate differences induced through temperature changes,
- downstream defect-history changes after thermal feedback is coupled back into Module 3,
- added computational cost.

A useful rule is that the ballistic-diffusive module should earn its additional complexity by changing one or more project-level quantities of interest enough to matter.

11 Module interfaces in the new 2D plan

11.1 Module 1 to Module 3 handoff

The first strong interface should now be a 2D deposition or defect-source map accompanied by interaction-resolved metadata. A minimal handoff object should contain coordinate arrays, a deposited-energy density field, optional time-resolved source information, interaction-type tallies or summaries, secondary-particle diagnostics when available, and metadata describing the incident particle, targeted energy band, device geometry, physics list, bin width, radiation case, and units.

11.2 Module 3 to Module 2 handoff

After a defect-evolution run, MATLAB should save at minimum coordinate arrays, effective defect fields, charged-defect density fields, trap-density fields, and time-history diagnostics for provenance.

11.3 Module 2 and Module 4 to Module 5 handoff

Carrier transport should read electric potential $\phi(x, y)$, electric-field components $E_x(x, y)$ and $E_y(x, y)$, temperature $T(x, y)$, and trap or recombination parameters derived from the defect state.

12 Verification strategy under the 2D architecture

Each module needs at least one low-complexity verification path.

Module	Verification logic

1	Compare deposition trends against known silicon behavior and verify that the exported 2D binning is internally consistent.
2	Use manufactured or symmetric charge configurations whose 2D potential and field patterns are physically obvious; verify sign conventions and boundary-condition behavior.
3	Check pure diffusion, pure annealing, and symmetry-preserving source cases in 2D. The structured-grid path and FEM path should both conserve total inventory under pure diffusion with zero-flux boundaries and reproduce the first-order annealing limit.
4	Recover expected limits of the 2D ballistic-diffusive thermal model, including equilibrium preservation, localized-pulse spreading, and measurable deviations from a Fourier baseline when boundary-memory effects matter; formal diffusive-limit comparison remains the next validation step.
5	Verify defect-free transport first, then turn on simple trap-assisted recombination. Check uniform no-field preservation, pure lifetime recombination, Gaussian diffusion inventory conservation, and current-sign conventions under a uniform electric field.
6	Confirm that decoupled or weakly coupled limits reduce to standalone solutions. Track iterative residuals and relaxation behavior.
7	Every scalable-prediction method must be benchmarked against a higher-fidelity reference. The benchmark should report both speedup and error metrics.
11	Recover known directed edges, relation types, strengths, lags, and SCC feedback groups in analytic and synthetic multiphysics systems; test direct-versus-indirect dependence, interventions, OOD mechanisms, calibration, conservation, and hidden-mechanism uncertainty.

13 Module 7: multiscale extrapolation and scalable prediction

A stronger title than "scaling up methods of the simulation" is **Multiscale Extrapolation and Scalable Prediction**. This module has three important branches:

1. **Controlled reduced-fidelity approximation:** coarser meshes, simplified defect networks, operator splitting, or reduced-order models. These must be judged by both speedup and error relative to a higher-fidelity reference.
2. **Statistical approximation and upscaling:** using smaller simulated tiles, event statistics, or surrogate source distributions to infer larger-scale behavior.
3. **Hybrid multiresolution coupling:** high fidelity only in physically critical regions, with

coarser or cheaper models elsewhere.

A useful project-level trade plot is approximation error versus speedup rather than speed alone.

14 Module 11: physical dependency graph proof-of-concept plan

Module 11 is restricted to the directed physical-variable graph. Its first implementation should proceed in the following order:

1. freeze the variable-node list, edge taxonomy, candidate directed-edge mask, sign priors, and allowed lag classes;
2. expose key Module 2–5 mechanisms through explicit graph switches and continuous edge multipliers;
3. generate constrained physical parameters, sources, initial conditions, and multiple graph families;
4. create matched hard, soft, node-clamp, and source interventions;
5. extract variable-node features from raw physical trajectories without using spatial adjacency as graph topology;
6. establish correlation, conditional, Granger-style, sparse-regression, MLP, fixed-graph, and NRI baselines;
7. train a directed typed message-passing network for state prediction, edge existence, relation type, strength, sign, lag, and uncertainty;
8. identify strongly connected components and topologically order the condensation graph to obtain a physical coupling sequence;
9. test noise, missing variables, held-out graph combinations, and hidden-mechanism discrepancy.

Synthetic data are sufficient to prove recovery of mechanisms represented in the generator. They are not sufficient to prove that the recovered graph contains all mechanisms present in a real irradiated device. Numerical solver order, block partitioning, relaxation, and runtime optimization are outside this module.

15 When this project-plan document should be updated

This document should be revised whenever a module definition changes, the dimensionality assumption changes, a new top-level file or folder becomes part of the standard workflow, an inter-module file format changes, the run order changes, a major verification strategy changes, or Module 7 gains a concrete benchmark or scaling method.

16 Near-term next steps after this revision

1. Keep the top-level `README.md`, `project_plan.tex`, and `project_plan.pdf` synchronized whenever the project architecture changes.
2. Verify the structured-grid and FEM Module 3 code paths, the FEM Module 4 code path, and the new FEM Module 5 carrier-transport code path, on the user's MATLAB environment.
3. Completed first stage: implemented the Module 3 causal-PINN pure-annealing/Gaussian driver and exact pure-annealing regression test described in `module3_cPIN.tex`. Next add a matched ordinary-PINN mode, a longer Gaussian convergence test, and an imported Geant4-like map.
4. Completed: implemented the first 2D reduced ballistic-diffusive Module 4 structured-grid MATLAB path with uniform-equilibrium, localized-pulse, and boundary-heating cases.
5. Keep the Module 4 note, file map, and run-order documentation synchronized with both the structured-grid and FEM thermal paths.
6. Completed: added Module 2, Module 3, Module 4, and Module 5 FEM implementation notes and first reduced FEM solver paths.
7. Freeze the first Module 11 physical-variable node list, candidate directed-edge mask, relation taxonomy, and intervention definitions.
8. Build the first 500–2,000-case graph-family synthetic dataset with explicit mechanism switches and matched interventions.
9. Establish non-graph structure-learning baselines and a fixed-graph predictive baseline before training a directed typed GNN.
10. Decide the first benchmark quantity of interest for Module 7 so that future approximations can be measured against something physically meaningful.